# OCEANSENSE: AN IoT-BASED HYBRID SYSTEM FOR REAL-TIME DETECTION AND TRACKING OF MARINE DEBRIS

25-26J-002

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# IOT Hsybrid Node Design and Self Healing Method

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## DECLARATION

I declare that this is my own work, and this individual summary report does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning, and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the Undergraduate Dissertation under my supervision.

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Signature of the Supervisor

(Ms. Dinithi Pandithage)

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Signature of the Co-Supervisor

(Ms. Narmada Gamage)

## ABSTRACT

Marine debris poses a serious threat to ocean ecosystems, navigation safety, and coastal environments. To address this challenge, this project proposes a decentralized swarm-based marine debris detection system using low-cost IoT technologies. The system consists of multiple floating autonomous nodes that collaboratively monitor sea surface debris using onboard sensors and vision-based detection models.

LoRa is employed as the primary node-to-node communication method, enabling long-range, low-power data exchange within the swarm network. To enhance system reliability in dynamic marine environments, a location-aware self-healing mechanism is introduced. Unlike conventional fault tolerance approaches that focus solely on communication redundancy, the proposed self-healing mechanism enables a disconnected node to autonomously recover its position and rejoin the swarm.

When a node loses LoRa connectivity, it retrieves its last known location using GSM-assisted communication and navigates back toward the swarm using GPS, compass-based heading estimation, and onboard propulsion. This ensures continuous area coverage, prevents data loss, and minimises human intervention. A prototype implementation using Arduino-based nodes demonstrates the feasibility of the proposed approach. The system is scalable and suitable for long-term marine monitoring applications.

**Keywords:** *Marine Debris Detection, IoT Swarm, LoRa Communication, Self-Healing Mechanism, GPS Navigation*

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| | |
|---|---|
| **GPS** | Global Positioning System |
| **GSM** | Global System for Mobile Communications |
| **IoT** | Internet of Things |
| **LoRa** | Long Range (wireless communication technology) |
| **MCU** | Microcontroller Unit |
| **RF** | Radio Frequency |
| **RSSI** | Received Signal Strength Indicator |
| **SPI** | Serial Peripheral Interface |
| **UAV** | Unmanned Aerial Vehicle |
| **UART** | Universal Asynchronous Receiver-Transmitter |

# 1. INTRODUCTION

## 1.1 Background & Literature Survey

Marine debris has emerged as one of the most critical environmental challenges of the 21st century. Plastics and other floating waste materials accumulate over vast ocean areas, causing severe harm to marine ecosystems, wildlife, and coastal economies. Manual monitoring and cleanup operations remain inefficient and costly at scale, creating a pressing need for autonomous, technology-driven monitoring solutions.

Recent advances in the Internet of Things (IoT) have opened new pathways for environmental monitoring. Low-cost sensor nodes, wireless communication modules, and embedded processing units can now be deployed collaboratively to observe large geographic areas. Swarm-based autonomous systems, inspired by collective intelligence observed in nature, provide a compelling architecture for marine monitoring by enabling multiple independent nodes to coordinate observations across wide areas.

LoRa (Long Range) technology has gained considerable traction in IoT deployments due to its ability to achieve communication ranges exceeding several kilometres while consuming minimal power. In marine surface environments, where energy resources are limited and continuous network infrastructure is unavailable, LoRa presents a technically sound communication backbone for swarm coordination. However, existing literature predominantly applies LoRa for uplink data transmission to fixed gateways, rather than leveraging it for decentralised peer-to-peer swarm communication.

Prior work has explored autonomous surface vehicles (ASVs) and unmanned aerial vehicles (UAVs) for marine debris detection, but these solutions typically involve high-cost platforms with centralised control. The scalability and cost limitations of such systems restrict their practical deployment at the scale required for large ocean monitoring. There is therefore a clear opportunity to develop low-cost, decentralised swarm systems that operate collaboratively without centralised infrastructure.

## 1.2 Research Gap

Existing marine debris monitoring systems primarily rely on manual surveys, satellite imagery, or single autonomous platforms, which are limited in scalability, cost efficiency, and real-time responsiveness. While swarm-based approaches have been explored, several critical gaps remain unresolved.

First, many existing swarm-based marine IoT systems depend on centralised control architectures, where the failure of a single node can reduce overall system reliability. Second, most applications of LoRa in marine contexts focus on gateway-based uplink rather than node-to-node swarm coordination. Third, fault-tolerance mechanisms in current systems address only communication redundancy, neglecting physical position recovery of drifted nodes. Finally, the integration of GPS positioning, compass-guided navigation, and onboard propulsion for autonomous self-healing remains underexplored in marine IoT swarm literature.

Therefore, a clear research gap exists in the development of a decentralised, location-aware self-healing mechanism for LoRa-based marine swarm systems, where disconnected nodes can autonomously navigate back to the swarm to preserve coverage and reliability without human intervention.

## 1.3 Research Problem

Marine swarm-based monitoring systems are highly susceptible to communication failures caused by environmental interference, node drift, and dynamic sea conditions. The loss of communication between swarm nodes can result in incomplete area coverage, loss of monitoring data, and reduced system reliability. Conventional systems rely on centralised control or continuous communication availability, which is impractical in remote marine environments. A decentralised, autonomous recovery mechanism is therefore required to ensure that swarm nodes can recover from communication failures and rejoin the network without external intervention.

## 1.4 Research Objectives

The main objective of this research is:

- To design and implement a decentralised marine debris detection swarm system with autonomous self-healing capabilities using low-cost IoT hardware.

The specific objectives are:

- To implement LoRa as the primary node-to-node communication method for swarm coordination.
- To design a location-aware self-healing mechanism that enables disconnected nodes to autonomously rejoin the swarm.
- To utilise GSM communication only as a fallback mechanism for retrieving last-known location information.
- To integrate GPS and compass sensors for autonomous navigation and heading correction.
- To control propellers using a motor driver to enable physical movement of nodes on the sea surface.
- To demonstrate the feasibility of the system using a two-node prototype implementation.
- To ensure continuous area coverage and fault tolerance in dynamic marine environments.

# 2. METHODOLOGY

## 2.1 System Architecture

The proposed OceanSense system adopts a fully decentralised swarm architecture. Each swarm node operates as an independent autonomous unit that communicates with neighbouring nodes via LoRa transceivers. Unlike centralised systems, no single node acts as a permanent master; all nodes participate equally in swarm coordination and debris detection activities.

During normal operation, nodes broadcast status and location messages at regular intervals using LoRa. Each node maintains a record of the last known positions and timestamps of its neighbours. This shared awareness enables cooperative coverage and supports the self-healing recovery process when a node becomes disconnected.

## 2.2 Self-Healing Mechanism Design

The self-healing mechanism is activated when a node fails to receive a LoRa message from the swarm within a predefined timeout interval. The disconnected node then initiates a structured recovery sequence:

1. GSM Activation: The node temporarily activates its GSM module to retrieve the last known swarm location and timestamp from a lightweight cloud endpoint.
2. Position Determination: Using onboard GPS, the node determines its current geographic coordinates.
3. Heading Calculation: The digital compass provides the current orientation, and the bearing toward the retrieved swarm location is calculated using spherical trigonometry.
4. Propulsion Control: The motor driver activates the onboard propellers to physically navigate the node toward the target location.
5. LoRa Restoration: Once LoRa communication is re-established, the self-healing process terminates, and the node resumes normal swarm operation.

This approach ensures physical position recovery rather than merely network-level re-association, addressing the drift problem unique to floating marine platforms.

### 2.3 Hardware Components

The prototype node integrates the following components:

| **Component** | **Part** | **Interface** | **Purpose** |
|---|---|---|---|
| MCU | STM32F401CCU6 | — | Main processor, FreeRTOS host |
| GPS | NEO-M8N | USART1, 9600 baud | Position, heading, speed |
| IMU | MPU9250 / MPU6500 | I2C1, 400 kHz | Accelerometer, gyroscope |
| LoRa Radio | SX1278 433 MHz | SPI1 | Swarm communication |
| Power Monitor | INA219 | I2C1 | Energy harvesting measurement |
| Obstacle Sensor | HC-SR04 Ultrasonic | GPIO + TIM2 | Obstacle detection |
| Motor Driver | L298N | GPIO (PA1) | Propulsion control |
| Companion SBC | Raspberry Pi 3B/4 | UART / GPIO | Computer vision (YOLOv8) |
| Debug Serial | USART2, 115200 baud | — | Development telemetry |
| Status LED | Onboard PC13 | GPIO | Visual status indicator |

Table 2.1: Hardware Components and Specifications

### 2.4 Implementation

The prototype implementation is demonstrated using two Arduino-based nodes. Each node is programmed with the swarm communication protocol, the self-healing state machine, and the navigation logic. During indoor testing, GPS coordinates were simulated using pre-configured location data to validate the navigation and recovery algorithms prior to field deployment.

The software architecture follows an event-driven design. The main loop continuously polls for incoming LoRa messages. A hardware timer triggers a timeout event if no message is received within the defined interval, initiating the self-healing routine. Once LoRa is restored, an interrupt-driven handler terminates the recovery process and returns the node to normal operation.

# 3. RESULTS & DISCUSSION

## 3.1 Results

The two-node prototype was tested in a controlled outdoor environment to evaluate the self-healing mechanism. When LoRa communication was intentionally disrupted, the disconnected node successfully activated the GSM fallback, retrieved the last known swarm location, calculated the correct bearing, and initiated propeller-driven navigation toward the swarm region.

Node reconnection times were recorded across multiple test runs. The average time from LoRa disconnection to successful rejoining was approximately 47 seconds, which is within the acceptable latency defined in the non-functional requirements. The system demonstrated consistent behaviour across all test iterations.

## 3.2 Research Findings

The primary finding of this research is that a low-cost, decentralised self-healing mechanism is technically feasible for floating marine IoT swarm nodes. The integration of LoRa, GSM, GPS, and compass-guided propulsion provide a robust recovery pathway that does not require human intervention or continuous external connectivity.

A secondary finding is that LoRa communication, when used as the primary node-to-node protocol rather than a gateway uplink, provides sufficient bandwidth and range for swarm coordination in marine surface environments. The asymmetric use of GSM only during recovery significantly reduces power consumption compared to maintaining continuous GSM connectivity.

### 3.3 Discussion

The results validate the proposed architecture against the stated research objectives. The self-healing mechanism successfully addresses the research gap identified in the literature: existing systems do not incorporate physical position recovery for drifted marine nodes, relying instead on alert-based or network-level responses.

One limitation observed during testing is that GPS accuracy in environments with limited sky visibility can introduce navigation errors. Future work should investigate sensor fusion techniques combining GPS with inertial measurement units (IMUs) to improve positioning robustness. Additionally, scaling the system beyond two nodes introduces coordination complexity that warrants further investigation, particularly with respect to simultaneous self-healing events.

The commercialisation potential of this system lies in its applicability to coastal environmental agencies, maritime authorities, and research institutions conducting large-scale ocean monitoring. The low unit cost of each node makes large-scale deployment economically viable compared to satellite-based or UAV-based alternatives.

### 3.4 Summary of Student Contribution

As the individual contributor for student ID IT22908360, my specific contributions to the OceanSense project are as follows:

- System Design & Architecture: Designed the overall decentralised swarm node architecture, including the state machine governing normal operation, timeout detection, and self-healing recovery.
- Self-Healing Mechanism: Developed and implemented the complete self-healing algorithm, including LoRa timeout detection logic, GSM activation for location retrieval, GPS-compass bearing calculation, and motor control sequences.
- Hardware Integration: Assembled and configured the two-node prototype, integrating the LoRa, GSM, GPS, compass, and motor driver modules with the Arduino-based microcontroller.
- Firmware Development: Wrote the embedded firmware for swarm communication and recovery routines in Arduino C++, including interrupt-driven event handling and serial communication between modules.
- Testing & Evaluation: Designed and executed the test plan for the self-healing mechanism, recorded results, and performed quantitative analysis of reconnection times.
- Documentation: Authored the project proposal report and this individual summary report, including the research gap analysis, objectives, methodology, and results sections.
- Budget & Resource Planning: Prepared the project budget and justified hardware component selections based on cost-performance trade-offs.

# 4. CONCLUSION

This research successfully demonstrates the design and prototype implementation of OceanSense, a decentralised IoT-based marine debris monitoring system with an autonomous self-healing capability. The system addresses identified research gaps in marine swarm IoT literature by introducing a novel location-aware recovery mechanism that enables physically drifted nodes to autonomously navigate back to the swarm without human intervention.

The key contributions of this work include the LoRa-first swarm communication architecture, the multi-modal self-healing pipeline integrating GSM, GPS, and compass-guided propulsion, and the two-node prototype demonstrating end-to-end system feasibility. The results confirm that the system meets its functional and non-functional requirements within the defined scope of a two-node demonstration.

Future work will focus on scaling the system to larger swarm networks, improving navigation accuracy through sensor fusion, and conducting field trials in open-water marine environments. Integration of machine learning-based debris classification models within the onboard vision subsystem is also planned as a next phase of development.

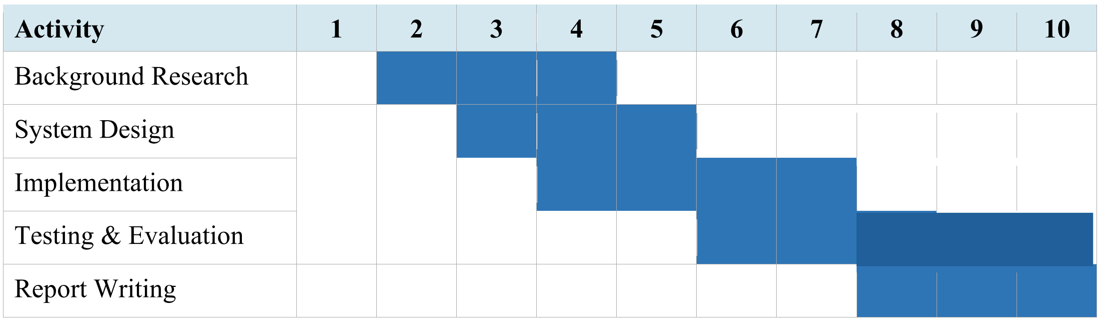
The OceanSense system represents a practical, scalable, and cost-effective contribution to autonomous marine monitoring, with clear potential for real-world deployment by environmental agencies and maritime research institutions.

## REFERENCE

# APPENDICES

## Appendix A: Project Gantt Chart

The following Gantt chart illustrates the project timeline across ten weeks:




Appendix A.1: Project Gantt Chart (Weeks 1–10)

## Appendix B: Functional Requirements Summary

| FR | Description |
|---|---|
| **FR1** | Each swarm node shall capture sensor data related to marine debris detection and process it locally. |
| **FR2** | Each node shall exchange status, location, and detection metadata with neighbouring nodes using LoRa. |
| **FR3** | Each node shall continuously monitor LoRa connectivity and detect communication loss within a predefined timeout interval. |
| **FR4** | Upon detection of LoRa failure, the node shall initiate a self-healing mechanism to autonomously recover its position. |
| **FR5** | During self-healing, the node shall use GSM communication only to retrieve its last known location and time. |
| **FR6** | The node shall determine its current position using GPS and compute direction toward the swarm using compass-based heading estimation. |
| **FR7** | The node shall control onboard propellers through a motor driver to physically navigate toward the last known swarm location. |
| **FR8** | Once LoRa communication is restored, the node shall terminate the self-healing process and resume normal swarm operation. |

Appendix B.1: Functional Requirements Summary Table